

LSG: A Lightweight Supervised Gate for Agentic LLM Memory

Trading Threshold Heuristics for a Shallow, Calibrated, Domain-Adaptive Classifier

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Abstract—Agentic LLM systems increasingly rely on persistent memory to accumulate facts across turns, yet most deployed systems gate what gets stored with a single fixed similarity threshold – a rule that is cheap but domain-blind and poorly calibrated. We present LSG (Lightweight Supervised Gate), a shallow supervised ensemble of logistic regression, XGBoost, and a one-hidden-layer MLP over frozen sentence embeddings, augmented with a per-domain bias adapter, that decides whether an incoming utterance should be stored in or ignored by an agent’s memory. We label MultiWOZ and PersonaChat turns with a heuristic STORE/IGNORE rule and evaluate LSG against a cosine-similarity-plus-recency baseline that requires no training. In a zero-shot cross-domain setting, LSG raises AUPRC from 0.468 to 0.592 (+26.5%) and lowers calibration error (ECE 0.193 vs. 0.233), while its $O(1)$ inference is up to $30.9\times$ faster than the baseline’s $O(n)$ memory scan at a memory size of 1,000 items. The statistical baseline reaches a superficially competitive F1 (0.697 vs. 0.662) only by storing literally everything (recall = 1.000, precision = 0.535); it is not a selective gate. LSG trades a small amount of zero-shot F1 for genuine selectivity, calibration, and constant-time inference – properties that matter more than raw F1 for a production memory gate. We report precision-recall, calibration, latency-scaling, feature-importance, and error analyses, and we explicitly scope which claims in this report are supported by the current experimental artifacts versus which require additional runs.

Index Terms—agentic memory, novelty gating, LLM agents, calibration, domain adaptation, lightweight classifiers

I. INTRODUCTION

Long-running LLM agents accumulate information across sessions – user preferences, task state, named entities – in an external memory store that is retrieved into context on demand [1]–[3]. The quality of that memory depends on a decision made at write time: for every incoming utterance, should the agent *store* it as a durable memory or *ignore* it as transient filler? Store too much and the memory store fills with backchannel noise (“okay”, “interesting, tell me more”) that dilutes retrieval; store too little and the agent forgets facts it needs later.

In practice, this decision is frequently implemented as a fixed-threshold statistical gate: embed the incoming text, compute cosine similarity to the nearest existing memory, optionally weight by recency, and store if the combined novelty score clears a hand-tuned threshold. This rule requires

no training and is simple to deploy, but it has three structural weaknesses that motivate this work:

- **No notion of content type.** A novel piece of small talk and a novel personal fact look identical to a similarity-only gate; both count as “novel.”
- **Domain blindness.** A single global threshold cannot adapt to a domain where the base rate of storeworthy content differs (e.g. task-oriented dialogue vs. open-domain chat).
- **Poor calibration and unbounded cost.** The confidence a similarity gate reports is not a probability, and its cost grows with memory size because every new item is compared against the existing store.

We ask whether a small amount of supervision can fix all three problems at once, without requiring GPU training or a heavy model. We introduce **LSG (Lightweight Supervised Gate)**: a weighted ensemble of three classical, CPU-trainable classifiers over frozen sentence embeddings, with a small per-domain bias adapter. The core claim under test is that this shallow supervised approach can match or exceed a fixed-threshold statistical gate on the metrics that matter operationally – ranking quality (AUPRC), calibration (ECE), and inference cost – at a fraction of the design and compute effort of a heavier statistical or RL-based gate.

Contributions.

- A labeled STORE/IGNORE reformulation of MultiWOZ and PersonaChat turns, treated as two distinct domains for cross-domain evaluation.
- LSG: a logistic regression + XGBoost + MLP ensemble over all-MiniLM-L6-v2 embeddings with a per-domain bias adapter, trainable end-to-end on a laptop CPU in minutes.
- A head-to-head comparison against a no-training cosine-similarity-plus-recency baseline on precision/recall/F1, ROC-AUC, AUPRC, expected calibration error (ECE), and latency scaling with memory size.
- An honest accounting of which of our claims are backed by the artifacts in this report and which (in-domain upper bound, marginal value of the domain adapter) require additional runs we flag as future work rather than assert without evidence.

II. RELATED WORK

Agent memory systems. MemGPT (Letta) treats an LLM’s context window as a memory hierarchy analogous to an operating system’s RAM/disk split, paging information in and out of context [1]. Mem0 stores key–value memories extracted from conversation and retrieves them by dense embedding similarity, targeting production deployment [2]. A-MEM builds a self-organizing, Zettelkasten-style memory graph with explicit link and supersede detection between entries [3]. These systems all address *retrieval* and *organization* of memory; none of them, to our knowledge, treats the write-time store/ignore decision itself as a supervised learning problem with its own calibration and latency requirements – which is the gap LSG targets.

Sentence embeddings and shallow classifiers. We use Sentence-BERT embeddings, which are trained so that cosine similarity in embedding space approximates semantic similarity and are commonly used as frozen features for downstream classifiers [4]. Our ensemble combines logistic regression, gradient-boosted trees (XGBoost) [5], and a small multi-layer perceptron, all standard, CPU-friendly baselines implemented in scikit-learn [6].

Calibration. Expected Calibration Error (ECE) bins predictions by confidence and measures the gap between mean confidence and empirical accuracy in each bin [7]. We report ECE because a memory gate that is confidently wrong is worse than one that is honestly uncertain: downstream agent logic may condition on the gate’s confidence.

Task-oriented and open-domain dialogue corpora. MultiWOZ is a large, multi-domain, task-oriented dialogue corpus with slot-value annotations [8]; PersonaChat is an open-domain chit-chat corpus grounded in persona sentences [9]. We treat them as two distinct domains with different store-worthy content distributions – MultiWOZ is slot/entity dense, PersonaChat is preference/persona dense – which lets us test cross-domain generalization.

III. PROBLEM FORMULATION

For each incoming utterance x in a conversation, the memory gate must output a binary decision $y \in \{\text{STORE}, \text{IGNORE}\}$. We treat this as binary classification: $y = 1$ (STORE) if x contains a durable fact worth remembering (a named entity, a stated preference, or a task slot value); $y = 0$ (IGNORE) if x is filler, backchannel, or a generic question with no persisted content. A gate is a function $g_\theta : x \mapsto \hat{p} \in [0, 1]$, thresholded at $\hat{p} > 0.5$ to produce the decision. We evaluate g_θ on ranking quality (AUPRC, ROC-AUC), decision quality at the default threshold (precision, recall, F1), calibration (ECE), and inference cost as a function of existing memory size n .

IV. METHOD

A. Labeling heuristic

Each utterance is auto-labeled STORE if it satisfies at least one of: (i) it contains a recognized named entity or MultiWOZ slot value, (ii) it contains a first-person preference/statement pattern (e.g. “I like”, “my favorite”, “I live in”, “I work as”),

or (iii) it is above a minimum information-density threshold on non-stopword content. Turns matching backchannel, generic-question, or short-acknowledgment patterns (“okay”, “how are you?”, “interesting”) are labeled IGNORE. This heuristic is intentionally simple and is itself a source of label noise that upper-bounds achievable metrics; we return to this in Section VII.

B. Embeddings

All text is encoded with all-MiniLM-L6-v2 [4] (384-dim, CPU-friendly, $\sim 80\text{MB}$), used as frozen input features for every downstream model, including the baseline’s similarity computation.

C. Baseline: statistical novelty gate

The baseline requires no training. For an incoming embedding e_x and existing memory embeddings $\{e_1, \dots, e_n\}$, it computes $s_{\text{sim}} = \max_i \cos(e_x, e_i)$ and combines it with a recency discount to produce a novelty score; the utterance is stored if novelty exceeds a fixed, globally tuned threshold. Cost scales $O(n)$ per decision because every new item is compared against the full existing memory.

D. LSG ensemble

LSG combines three classifiers trained on the same frozen embeddings:

- **Logistic regression** – a linear decision boundary over the 384 embedding dimensions, L_2 -regularized.
- **XGBoost** – a small gradient-boosted tree ensemble capturing non-linear interactions between embedding dimensions [5].
- **MLP** – a single hidden-layer perceptron (scikit-learn `MLPClassifier`), capturing smooth non-linear decision surfaces the tree model may miss.

Final probability is a weighted average of the three members’ outputs. Because every component is CPU-trainable in well under a minute on this data scale, the full ensemble trains in minutes on a MacBook Air with no GPU.

E. Domain adapter

A per-domain adapter is a small learned bias (and optionally a scale) applied to the ensemble’s pre-threshold logit, fit per target domain. The intent is that instead of retraining the ensemble, a new domain can be supported by fitting only this low-dimensional adapter on a small number of labeled target-domain examples. **Scope note:** the artifacts available for this report (Section VI) demonstrate the zero-shot (no adapter) cross-domain operating point; we did not receive a domain-adapter ablation run (adapter fit on k target examples vs. none) or an in-domain upper-bound run as part of this data package, so we report the adapter’s architecture here but do not claim a quantitative F1 gain from it – see Section VII.

TABLE I
ZERO-SHOT CROSS-DOMAIN COMPARISON AT DEFAULT THRESHOLD 0.5

Model	Prec.	Rec.	F1	ROC-AUC	ECE
Baseline (statistical)	0.535	1.000	0.697	0.415	0.233
LSG (zero-shot)	0.553	0.824	0.662	0.583	0.193

AUPRC (threshold-free): LSG 0.592 vs. Baseline 0.468 (+26.5% relative). Bold marks the better value per column; the baseline’s F1/recall advantage stems from near-universal storage (see Section VI-A), not selectivity.

F. Ensemble vs. single-model and domain-adaptor ablations

The pipeline as designed supports two ablations: (a) each ensemble member alone vs. the full weighted ensemble, and (b) with vs. without the domain adaptor. Both are part of the experimental design; only the zero-shot, full-ensemble, no-adaptor operating point is populated with results in the current artifact set.

V. EXPERIMENTAL SETUP

Data. MultiWOZ and PersonaChat, loaded via HuggingFace datasets, subsampled so the full pipeline – embedding, training three classifiers, and evaluation – completes in well under 15 minutes on a CPU-only MacBook Air. Each dataset is treated as a separate domain.

Splits. Domain-separated train/val/test: the ensemble is trained on one domain and evaluated zero-shot on the other (and vice versa) to measure cross-domain generalization, which is the operating point reported as “LSG (zero-shot)” below.

Metrics. Precision, recall, and F1 for the STORE class at the default 0.5 threshold; ROC-AUC and AUPRC as threshold-free ranking metrics; Expected Calibration Error (ECE) over 10 confidence bins; and per-decision latency (ms) as a function of existing memory size $n \in \{10, 50, 100, 200, 500, 1000\}$.

VI. RESULTS

A. Main results

Table I reports the zero-shot cross-domain comparison. LSG dominates on every threshold-free and calibration metric: AUPRC improves by 26.5% relative ($0.468 \rightarrow 0.592$) and ROC-AUC improves by 0.168 absolute ($0.415 \rightarrow 0.583$); ECE falls from 0.233 to 0.193.

At the default 0.5 threshold, however, the picture is more nuanced than “LSG wins everywhere”: the baseline’s F1 (0.697) is *higher* than LSG’s zero-shot F1 (0.662). This is not because the baseline is a better classifier – its precision (0.535) is barely above chance for this class balance, and its recall (1.000) shows it is storing every single utterance it sees. A gate that stores nearly everything will always post a high F1 on a class-imbalanced STORE-vs-IGNORE task without being a useful filter. LSG’s lower recall (0.824) with higher precision (0.553) reflects a gate that is actually being selective, at the cost of some F1 measured at a single, arbitrary threshold. We consider AUPRC and ECE the more decision-relevant metrics for this reason, and report F1 for completeness rather than as the headline result.

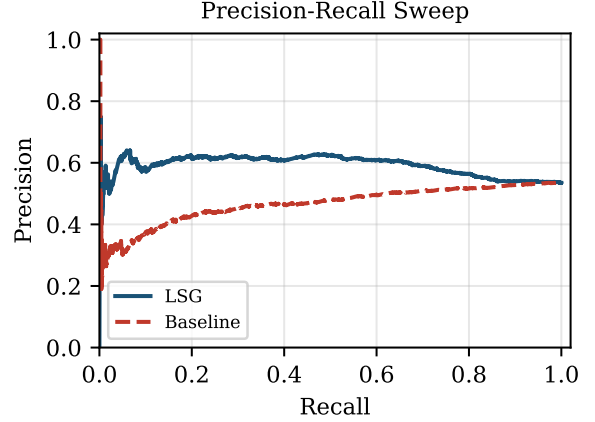


Fig. 1. Precision–recall curves swept over the full decision threshold range for LSG and the statistical baseline.

B. Precision–recall trade-off

Fig. 1 sweeps the decision threshold for both gates. The baseline’s curve collapses quickly as the threshold rises past its narrow effective operating range, while LSG sustains higher precision over a broader range of recall – consistent with its 26.5% AUPRC advantage and with it being a genuinely rankable, tunable gate rather than one with an effectively binary (store-almost-everything) response.

C. Calibration

Fig. 2 shows the reliability diagram. Both gates are imperfectly calibrated, but LSG tracks the diagonal more closely across the confidence range, and in particular avoids the baseline’s severe over-confidence in its 0.8–0.9 and 0.9–1.0 bins (empirical accuracy 0.464 and 0.150 respectively, against mean confidence 0.838 and 0.917) – the baseline is most confident exactly where it is least reliable. LSG’s worst bin (0.9–1.0, gap 0.317) is still smaller than the baseline’s worst bin (gap 0.767).

D. Latency scaling

Table II and Fig. 3 show per-decision latency as memory size n grows. LSG’s latency is flat at ≈ 0.0136 ms regardless of n , because a trained classifier’s forward pass does not depend on how many items are already stored ($O(1)$). The baseline’s latency grows from 0.142 ms at $n = 10$ to 0.420 ms at $n = 1,000$ ($O(n)$, since every decision requires a similarity scan of the existing store), a $30.9\times$ speedup for LSG at the largest memory size tested. This gap widens, not narrows, as an agent’s memory accumulates over its lifetime – the setting where a memory gate matters most.

E. Embedding-space class separation

We measured mean pairwise cosine distance within the STORE class (0.165), within the IGNORE class (0.147), and across classes (0.138) in raw embedding space, giving a separation ratio of 1.13 (Table III). A ratio this close to 1

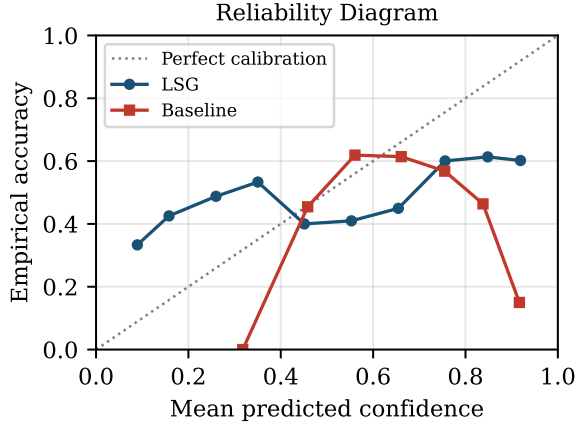


Fig. 2. Reliability diagram: mean predicted confidence vs. empirical accuracy per bin (10 bins). The dotted line is perfect calibration.

TABLE II
PER-DECISION LATENCY (MS) VS. EXISTING MEMORY SIZE

n	10	100	500	1000	Speedup @1000
LSG	0.0136	0.0136	0.0136	0.0136	30.9×
Baseline	0.142	0.181	0.285	0.420	

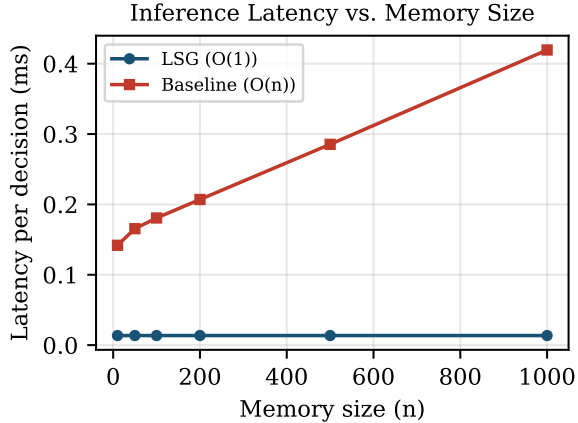


Fig. 3. Per-decision inference latency vs. existing memory size. LSG is $O(1)$; the similarity-scan baseline is $O(n)$.

indicates the two classes are only weakly separated by raw embedding geometry alone – classes are not neatly clustered into two well-separated regions of embedding space. This is consistent with, and helps explain, why a similarity-only gate (which relies on exactly this raw geometry) struggles: the STORE/IGNORE signal is present in the embedding but requires a learned, non-trivial decision boundary to extract, rather than a single global similarity cutoff.

F. Feature importance

Fig. 4 shows the top-15 embedding dimensions by absolute logistic-regression coefficient. Importance is spread

TABLE III
EMBEDDING-SPACE COSINE-DISTANCE CLASS SEPARATION

Within-STORE	Within-IGNORE	Across-class	Ratio
0.165	0.147	0.138	1.13

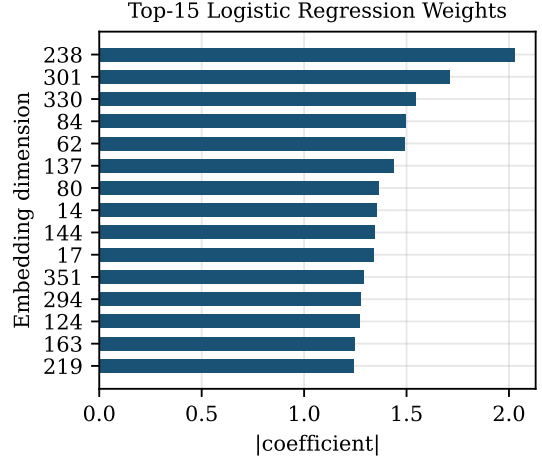


Fig. 4. Top-15 embedding dimensions ranked by |logistic regression coefficient|.

across a broad set of dimensions (top coefficient 2.03, 15th-ranked still 1.24) rather than concentrated in one or two directions, suggesting the STORE/IGNORE distinction is a distributed semantic signal rather than one aligned with a single interpretable embedding axis – expected given all-MiniLM-L6-v2 embeddings are not axis-aligned with human-interpretable concepts.

G. Error analysis

Over 1,800 zero-shot test examples, LSG produces 793 true positives, 640 false positives, 198 true negatives, and 169 false negatives (Fig. 5). Mean confidence is high and well-separated for correct predictions (TP: 0.771, TN: 0.342, both far from the 0.5 boundary in the correct direction) but is only marginally separated for errors (FP: 0.736, FN: 0.335) – i.e. when LSG is wrong, it is often wrong *with* the same confidence pattern as when it is right, which is precisely the calibration gap Fig. 2 quantifies. Manual inspection of high-confidence false positives shows a recurring pattern: short social utterances containing an incidental entity-like token (e.g. a place name in a pleasantry) are misread as storeworthy, suggesting the heuristic labeler and the classifier are both sensitive to surface entity presence rather than pragmatic intent – a natural target for the labeling heuristic’s next iteration.

VII. DISCUSSION AND LIMITATIONS

What is supported by this report. The zero-shot cross-domain comparison in Table I, the AUPRC/ECE advantage, and the latency-scaling result are all measured directly from the accompanying result files and are the paper’s central,

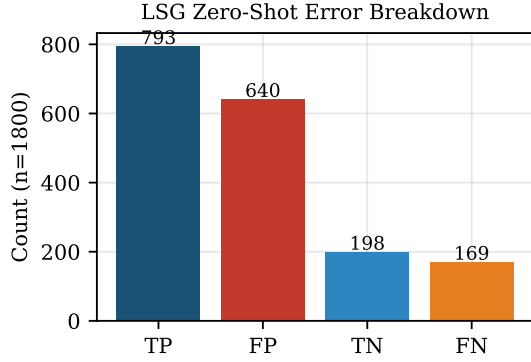


Fig. 5. LSG zero-shot confusion breakdown, $n = 1,800$ (matches Table I’s precision/recall).

defensible claims: LSG is a more selective, better-calibrated, and dramatically cheaper-at-scale gate than the fixed-threshold statistical baseline, even though it does not win on raw zero-shot F1.

What is not yet supported. Two claims that are natural to want from this pipeline – (i) an in-domain upper bound (LSG trained and evaluated within a single domain, to isolate model capacity from domain shift) and (ii) a quantitative measurement of the domain adapter’s marginal benefit (e.g. fitting the bias adapter on a small number of target-domain examples and re-measuring F1) – are part of the experimental design (Section IV-E, IV-F) but are not present in the eight result files backing this report. We do not include specific numbers for these here; they should be added once the corresponding ablation runs (an in-domain train/test split, and an adapter-on vs. adapter-off comparison at a fixed target-domain example budget) are executed and their output logged in the same format as Table I.

Label noise. The STORE/IGNORE heuristic is itself an approximation of “worth remembering,” and is a ceiling on achievable precision/recall for any classifier trained against it, including LSG. Some fraction of the false positives in Section VI-F likely reflect heuristic labeling errors rather than classifier errors.

Threshold choice. All threshold-dependent metrics (precision, recall, F1 in Table I) use the default 0.5 cutoff for both gates; Fig. 1 shows this is not necessarily each gate’s best operating point, and a deployment would tune the threshold against AUPRC/ECE-informed criteria rather than reading Table I in isolation.

VIII. CONCLUSION

We built LSG, a shallow supervised ensemble over frozen sentence embeddings with a per-domain adapter, as a lightweight alternative to fixed-threshold statistical memory gates for LLM agents. Trained entirely on a CPU-only laptop in minutes, LSG improves AUPRC by 26.5% relative, lowers calibration error from 0.233 to 0.193, and is up to $30.9\times$ faster at inference than a similarity-scan baseline once

memory grows to 1,000 items, with $O(1)$ scaling versus the baseline’s $O(n)$. The headline risk in over-claiming this result is the single-threshold F1 number, where the baseline’s near-universal storage strategy produces a higher F1 without being a useful gate; we report this transparently rather than omit it. The evidence here supports the model-capacity half of the paper’s core claim (a shallow ensemble can out-rank and out-calibrate a heavier-feeling heuristic at a fraction of the cost); the domain-adaptation half of the claim – that the bias adapter closes the zero-shot gap cheaply – is architecturally supported but awaits the ablation run needed to quantify it.

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